

[7] Requirements

Project	Simplified electronic project designed to verify practical skills in electrical engineering, PCB design, electronic assembly, embedded software implementation, and programming within the Microchip Technology development environment. Simple powering.	MEET	The project is a simple LED light effect device with a user interface realized using a capacitive touch sensor. Powered by USB-C Connector.
MAIN OBJECTIVE			

1	The project shall contain very small integrated circuits. Pitch less than 1mm	MEET	IC: Cap sensor : AT42QT1012 : SOT-23-6 package : PITCH: 0.95 IC: uC : Attiny-204 : SOIC-14_3.9x8.7mm_P1.27mm : PITCH 1.27
2	The project shall contain 0805 SMT passives components.	MEET	Device contain multiple 0805 SMT elements.
3	The project shall contain multi-pin integrated circuits to verify soldering precision.	MEET	IC: Shift register : STP16CP05MTR : SO-24 : 24 pins IC: uC : Attiny204 : SOIC-14_3.9x8.7mm : 14 pins
4	The project shall contain an UPDI programming interface for firmware upload and debugging.	MEET	uC Attiny204 with UPDI programming interface
5	Possibility to program using Microchip enviroment and pickit-4 programmer.	MEET	It is possible to program the device using pickit-4 and MPLAB software.
6	Shall contain at least two status led.	MEET	Status LED'S : LED1, LED2.
7	Shall be able to interact with user.	MEET	The user can change the light effect using a capacitive touch sensor.
8	Shall be easy to power up.	MEET	Powered by USB-C connector with general mode
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		Comments:	Company: BaJRan imgaginary industries		Variant: CHECKED	Git Hash: ff91115
			Board Name: 3ART13 LED		Project Name: Sienkiewicz	
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